

Unanticipated Lessons from Communities: Navigating Society-Centered Research in the AI Era

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Abstract

As AI technologies increasingly integrate into daily life, their deployment often overlooks the complexities of the communities they aim to serve. This gap is particularly acute for marginalized communities, where AI can exacerbate inequalities due to technosolutionism – the tendency to frame technology as a one-size-fits-all solution. This Special Interest Group (SIG) will explore unexpected lessons from community-led AI initiatives, emphasizing strategies for meaningful collaboration, shared ownership, and equitable partnerships. Through discussions and storytelling, the SIG aims to advance community-centered approaches to AI development and foster a robust, sustained network of researchers and practitioners.

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CHI EA '25, Yokohama, Japan

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ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3716290>

CCS Concepts

• **Human-centered computing** → HCI design and evaluation methods.

Keywords

Society-Centered AI, Participatory Methods, Community-based Research

ACM Reference Format:

Ding Wang, Remi Denton, Anoop K. Sinha, Shruti Sheth, Lauren Wilcox, Maryam Mustafa, Motahhare Eslami, Ken Holstein, Maarten Sap, Angela D. R. Smith, Andrea G Parker, Neha Kumar, Naveena Karusala, Tawanna R Dillahunt, and Jared Lee Katzman. 2025. Unanticipated Lessons from Communities: Navigating Society-Centered Research in the AI Era. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3706599.3716290>

1 Introduction

Community-centered and participatory approaches have long been central to the field of Human-Computer Interaction (HCI), offering

unique benefits in designing technologies that address the needs and priorities of diverse communities. These methods enhance understanding of users and their contexts, foster a sense of ownership, increase the fit and adoption of technologies, and empower participants through collaborative processes [2, 5, 6]. Moreover, participatory approaches and community-collaborative research have demonstrated the potential to enhance lives, challenge social attitudes, and build equitable relationships between researchers and communities [8].

Recent critiques, however, have highlighted that participatory approaches, such as PD workshops, can reflect privileged and affluent practices that unintentionally neglect the historical and material contexts of underserved communities. For example, challenges such as limited community access, inequitable engagement with materials and activities, and potential harm when collecting personal narratives underscore the need for more equitable and context-sensitive design practices [7]. Similarly, research on community-based participatory approaches identifies key conditions for fostering equitable partnerships and power sharing, emphasizing their role in addressing systemic inequities and fostering mutual respect between researchers and communities [11].

The AI era introduces additional complexity to community-centered work, offering both new opportunities and challenges. While AI has unprecedented potential to address societal challenges, its development often risks marginalizing the very communities it seeks to serve. Techno-solutionism — the tendency to treat technology as a universal solution to complex social problems — can neglect the nuances of local contexts and exacerbate inequalities [9]. Participatory approaches are therefore critical to ensuring that AI systems align with the lived experiences, values, and goals of affected communities. Encouragingly, initiatives such as OpenAI’s Democratic Inputs to AI grant program¹ and Google’s awards for Collective & Society-Centered AI² have begun to support stakeholder engagement by funding community-centered research.

Despite the promise of participatory methods, the practical realities of community-centered work remain fraught with challenges. Power imbalances, ethical dilemmas, and resource constraints often make these efforts difficult to “get right.” Unexpected situations—such as difficulties in recruiting participants, managing relationships, or addressing community expectations—can derail projects or limit their impact. Conversations around these challenges often remain siloed, and the failures encountered in such work are rarely shared or systematically analyzed.

Building upon traditions of human-centered AI [10, 12], participatory AI [1, 3, 4], and community-collaborative research, this Special Interest Group (SIG) seeks to create a constructive and reflective space where researchers, practitioners, and community members can candidly share their experiences with community-centered AI and HCI projects, particularly when things have gone wrong. By exploring both successes and failures, the SIG aims to deepen understanding of the complexities of participatory work in the AI era. Topics will include addressing power dynamics, overcoming ethical and logistical challenges, and identifying strategies to co-create impactful and context-sensitive technologies.

¹<https://openai.com/index/democratic-inputs-to-ai-grant-program-update/>

²<https://research.google/programs-and-events/award-for-inclusion-research-program/>

Through facilitated discussions and knowledge exchange, this SIG will emphasize the importance of learning from unexpected situations, enabling participants to develop more resilient, equitable, and community-centered approaches to research and design. In doing so, it aims to build a community of practice that prioritizes transparency, mutual support, and collective learning.

2 Topics of Discussion

Participants will explore three core themes:

- **What Can Go Wrong in Community-Led AI:** Challenges such as power imbalances, unmet expectations, and risks of exacerbating inequalities. Ethical considerations and complexities in navigating community relationships.
- **Resisting Techno-Solutionism:** 1.) Strategies for co-defining problems with communities. 2.) Developing solutions that respect and integrate community knowledge. Ethical considerations and complexities in navigating community relationships.
- **Successful Community Collaborations and Best Practices:** 1.) Examples of trust-building and equitable partnerships. 2.) Lessons on how communities can take a leading role in defining projects.

3 Planned Activity

3.1 Format and Schedule

This 75-minute SIG session is designed to foster dialogue and community-building through structured discussions, storytelling, and collaborative planning.

- **Welcome and Session Goals (10 minutes):** The session begins with an introduction from the organizers, outlining goals such as reflecting on challenges, sharing lessons, and building strategies for community-led AI development. The introduction will emphasize the importance of centering community voices and ask a critical question – *What do communities want researchers and practitioners to learn from their experiences?* To set the stage, organizers will present brief vignettes or profiles of community partners, highlighting their contributions and perspectives.
- **Storytelling and Case Reflections (15 minutes):** Participants will hear short stories from researchers and community members sharing unexpected lessons, challenges, and successes from their experiences.
- **Thematic Breakout Groups (30 minutes):** Participants will divide into small groups to discuss themes aligned with the SIG’s objectives, including challenges, strategies for resisting techno-solutionism, and best practices for collaborations.
- **Group Report-Back and Collaborative Reflection (10 minutes):** Each group will share insights with the entire session to promote cross-group learning.
- **Continuity Plan Discussion and Feedback (5 minutes):** Organizers will present the continuity plan and seek feedback to refine ongoing initiatives.

- **Closing and Next Steps:** The session will conclude with a brief Q&A and a call to action, encouraging attendees to engage in the SIG's continuity efforts.

4 Continuity Plan

The SIG aims to establish a long-term network of researchers, practitioners, and community members dedicated to advancing community-led AI development. To ensure sustained engagement and alignment with community priorities, the following initiatives will be pursued:

- **Ongoing Mentorship and Peer Support:** A mentorship network will connect attendees with experienced researchers and practitioners. Early-career professionals and those new to community-led AI will receive tailored support to help them navigate challenges and develop effective, equitable practices.
- **Virtual Community Hub:** A centralized online platform will facilitate ongoing dialogue, knowledge exchange, and resource sharing. The hub will include case studies, best practices, and practical tools for creating equitable collaborations. It will also serve as a repository for SIG outcomes, such as shared vignettes and lessons learned.
- **Biannual Virtual Meetings:** Regular virtual meetings will sustain momentum and broaden participation, occurring twice a year to remain achievable. These sessions will feature guest speakers from academia, industry, and communities to discuss emerging trends, share practical strategies, and present case studies of successful community-led initiatives.
- **Community Feedback and Co-Design of Initiatives:** To ensure ongoing alignment with community priorities, the SIG will establish feedback loops with community partners. Community members will be invited to participate in periodic consultations, focus groups, or surveys to evaluate and refine SIG initiatives. This co-design approach will empower communities to shape future activities, ensuring that the SIG remains responsive to their needs and aspirations.
- **Using the CHI SIG Format for Continued Engagement:** The SIG will serve as a recurring event at CHI, providing an annual opportunity to convene researchers, practitioners, and community members in person. Each year at CHI, the SIG will build on past discussions and outcomes, creating continuity and ensuring sustained engagement with this community.
- **Collaborative Opportunities and Future Events:** The SIG will act as a springboard for future workshops and events, encouraging collaborative projects that engage diverse perspectives. Specific opportunities for co-creating ethical guidelines, shared resources, and community-led projects will be identified.
- **Engagement with Broader Ethical AI Initiatives:** Participants will be encouraged to connect with organizations such as SIGCHI ethics services to discuss challenges like funding transparency, equitable partnerships, and sustaining community relationships. The SIG will act as a channel for ongoing advocacy for community-centered practices in AI research and development.

By embedding community voices directly into its structure and fostering mentorship, resource sharing, and dialogue, this SIG aims to build a robust, sustained network. Future initiatives will prioritize empowering communities, amplifying their voices, and supporting the development of context-sensitive, meaningful AI solutions.

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